

Poncelet’s Time in Prison, Bezout’s Theorem, and a Porism Regarding Polygons

4:00 PM
April 28th, 2026
Roop 103

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Abstract

In this talk, we will discuss the history and construction of complex projective space via Poncelet, state (topological) Bezout’s theorem for smooth planar projective curves, and discuss Poncelet type theorems in the plane regarding in-and-circumscribed n -gons on pairs of smooth conics. Rather than focusing on technical proofs, we aim to provide a survey of interesting results in algebraic geometry which might inspire attendees to learn more about the subject.

Contents

1	Poncelet’s Time in Prison	2
2	Poncelet’s Porism	5
2.1	Group Actions on Elliptic Curves	6

1 Poncelet's Time in Prison

Say hello, thank Vlad for organizing, congratulate the other speakers etc.

Today, we are going to dive into some topics in algebraic geometry surrounding the theory of intersections of algebraic curves. Specifically, we are going to set up and walk through the solution to “Poncelet’s porism,” a problem proposed by Jean-Victor Poncelet in 1812, and published in 1814.

The math history here is neat - until Poncelet, we had no reason to use the complex numbers, due to the fact that no geometric construction could be made to justify $\sqrt{-1}$. However, after the Napoleonic wars (which Poncelet participated in as a military engineer for Napoleon), Poncelet was imprisoned for two years, and in those two years, (created and) solved three major issues in math:

1. Why we need the complex numbers for geometry,
2. Why we need “projective space” for geometry, and,
3. Poncelet’s Porism.

Towards this first point, since our goal is intersection theory, we really need to establish a consistent notion of the number of times which two curves intersect. Consider,

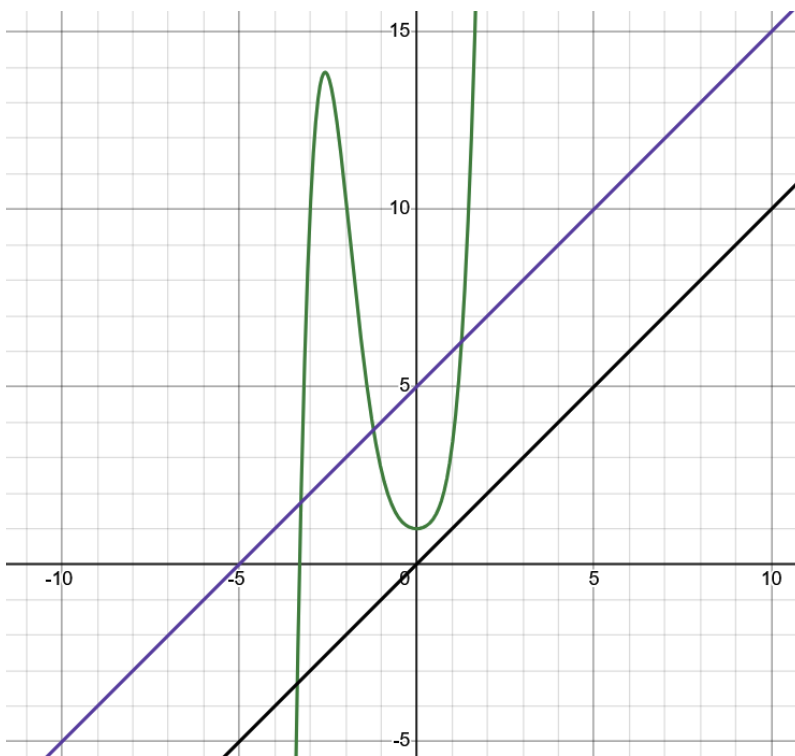


Figure 1: A Cubic and a Line

We know that the “number of bumps on a curve” should be the degree of the polynomial defining, i.e., the highest power of x in the polynomial. We can see that the maximum number of

intersections of a line with this curve is the degree of the curve, but if we vary the line, we certainly get fewer “real-valued” intersections. Poncelet showed that these actually show up in the complex plane! So, we want to build a parameter space where the number of intersections of two curves is entirely invariant of the position of the curve, avoiding specialized regions of the parameter space.

We would imagine that this should be $\mathbb{C}$, however, Poncelet also showed that the complex plane doesn't *quite* do the trick: it is a well known fact that polynomials with real coefficients and one complex root, $\alpha \in \mathbb{C} \setminus \mathbb{R}$ must have another complex root, $-\alpha$. From this, we see that **complex roots of polynomials come in pairs**. We also know by the fundamental theorem of algebra that a polynomial of degree n must have n complex roots (not necessarily of the form $a + bi$).

So, consider a polynomial of even degree with one $\mathbb{R}$ -valued intersection with a line. Where does the extra point happen? In these cases, it necessarily implies a division by zero - and so we would want our point of intersection to occur at infinity. But this cannot be well defined, right?

Let's consider another example:

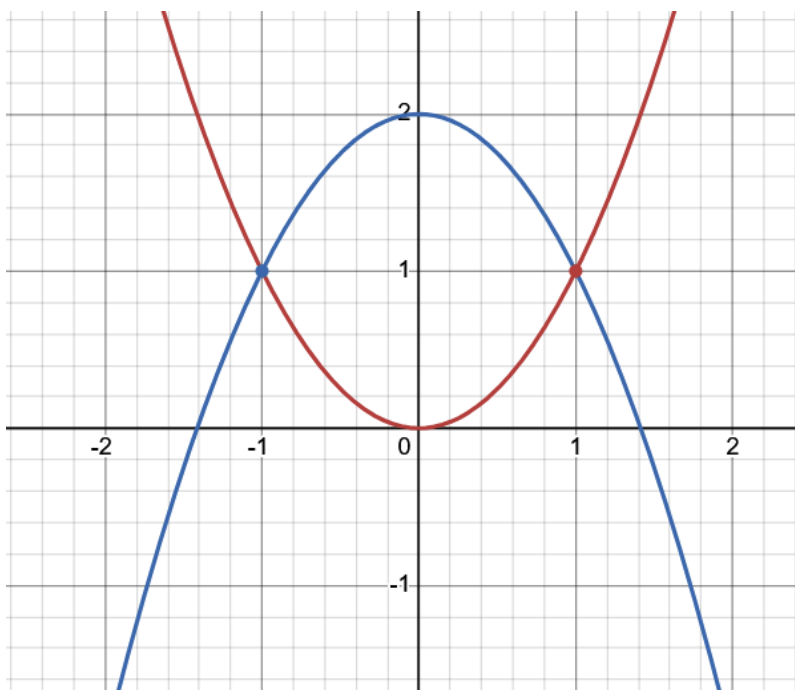


Figure 2: Parabolas Require $\mathbb{P}^2$

$$y = x^2$$

$$y = 2 - x^2$$

We can see that “in general position,” we should have four intersections:

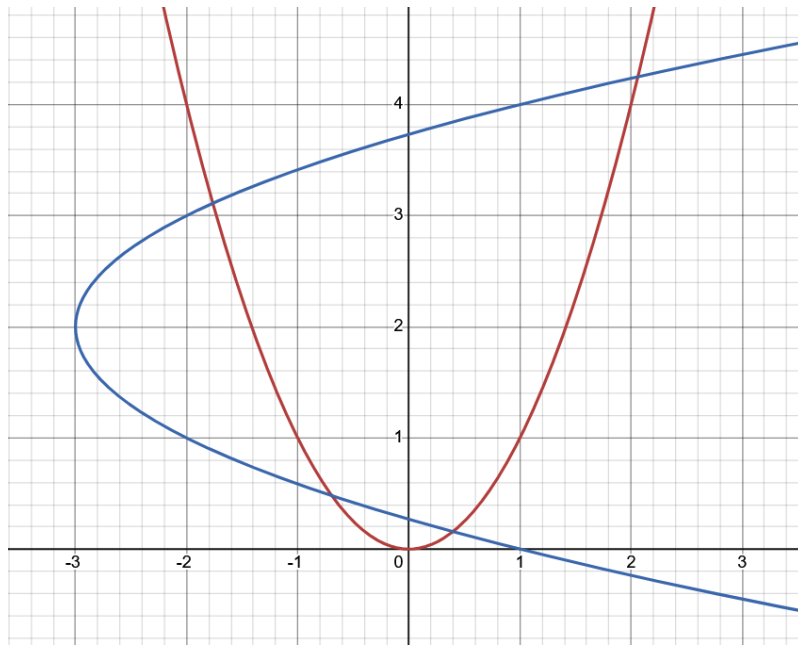


Figure 3: General Parabolas

So where do we place the other two intersections?

This requires the notion of *projective space*, or equivalently, a space where parallel lines meet at infinity. It is a bit complicated to explain why, but this is defined by

$$\mathbb{P}^n = \{\text{lines through the origin of } \mathbb{C}^{n+1}\}.$$

As a practical example, if we take our two parabolas and make the replacement,

$$\begin{aligned} x &\mapsto \frac{x}{z} \\ y &\mapsto \frac{y}{z}, \end{aligned}$$

then we get

$$\frac{y}{z} = \frac{x^2}{z^2} \Rightarrow yz^2 = x^2z \Rightarrow z(yz - x^2) = 0$$

and

$$\frac{y}{z} = 2 - \frac{x^2}{z^2} \Rightarrow yz^2 = 2z - x^2z \Rightarrow yz^2 + x^2z - 2z^3 = 0.$$

Without working out the gory details, it is clear that the following points satisfy

$$yz^2 - x^2z = yz^2 + x^2z - 2z^3.$$

Points:

- $[1 : 1 : 1]$
- $[-1 : 1 : 1]$

- $[1 : -1 : 0]$
- $[1 : -1 : 0]$.

So,

Theorem 1 (Bezout's Theorem). *For any two smooth, projective curves $f, g \subseteq \mathbb{P}^2$, of degree d, e respectively, $\#|f \cap g| = de$.*

Namely, projective space is the space where “geometry behaves consistently,” and Bezout's theorem holds for planar curves.

2 Poncelet's Porism

Now, we turn our attention towards Poncelet's theorem.

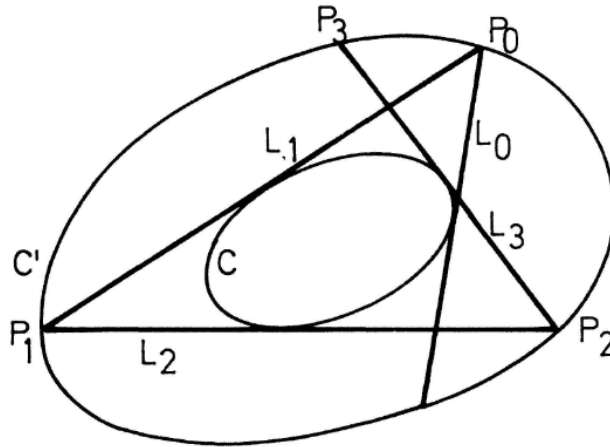


Figure 4: Poncelet's Porism

Consider two ellipses in the plane (these are really called “smooth conics”). If we pick a point p_0 on the outer ellipse, C_1 , we are given two options for tangent lines to the inner ellipse, C_2 . Without loss of generality, if we pick one of them, we can construct the tangent line T_0 and obtain another point, p_1 , on the outer ellipse. Now that we have only one choice for a new tangent line to C_2 , pick T_1 to go through p_1 and be tangent to C_2 . We may iterate this process; if we ever end up at p_0 again, we call the shape constructed by the T_i a polygon which is inscribed on C_1 and circumscribed on C_2 . Alternatively, we may refer to this as a in-and-circumscribed polygon on two smooth planar conics.

Theorem 2 (Poncelet's Porism). *There are either infinitely many in-and-circumscribed polygons on a pair of two smooth planar conics, or zero. Moreover, this result is entirely independent of the choice of starting point.*

Over the next few minutes, we will examine tractable elements of Harris and Griffiths' 1977 proof of this theorem.

To begin, we will establish the collection of all pairs of points and lines where the lines are tangent to C_2 and the points lie on both the lines and the outer circle, C_1 .

$$E = \{(P, L) \in C_1 \times C_2^* \cong \mathbb{P}^1 \times \mathbb{P}^{1*} \mid P \in L\}$$

It isn't explicitly clear from this setup, but this is an example of a mathematical object called a *genus one Riemann surface*. It is well beyond the scope of our talk to explain why, but all genus one Riemann surfaces are structurally the same - or isomorphic by the Abel-Jacobi map $J : E \rightarrow H^0(\Omega_E)/H_1(E; \mathbb{Z}) \cong \mathbb{C}/\Lambda$ - to another mathematical object called an elliptic curve.

Explanation for if people push on this for some reason: If we consider the map $E \rightarrow C_2^*$ given by $(P, L) \rightarrow L$, this realizes E as a 2-sheeted covering of the Riemann sphere, $\mathbb{P}^1$. Taking the intersection of C_1^* and C_2^* , we can see by Bezout's theorem that there should be four generic intersection points; these are the branch points of the covering, and occur for the lines tangent to both C_1 and C_2 in $\mathbb{P}^2$. So clearly, E is a nonsingular Riemann surface with Euler characteristic $\chi(E) = 2\chi(\mathbb{P}^1) - 4 = 0$, and so $g(E) = 1$.

This gives us a non-vanishing global differential form on E , which is unique up to scalars,

$$\int_{p_0}^p \omega.$$

Since the difference by scalar is determined by the winding number of the path $p_0 \rightarrow p$, or equivalently by the number of loops on $H^1(E, \mathbb{Z}) \subset H^1(\mathbb{C}, \mathbb{Z})$, we take the Abel-Jacobi map

$$J : E \rightarrow H^0(\Omega_E)/H_1(E; \mathbb{Z}) \cong \mathbb{C}/\Lambda$$

naturally. Namely, the elements of the image of J are such path integrals, where $b, p \in E$ map to

$$\int_b^p \omega \in H^0(\Omega_E)/H_1(E; \mathbb{Z}).$$

Thus, we have an analytic isomorphism of Riemann surfaces,

$$E \rightarrow J(E) \cong \mathbb{C}/\Lambda.$$

2.1 Group Actions on Elliptic Curves

Elliptic curves are equations of the general form,

$$E : y^2 = x^3 + ax + b.$$

It is an important fact to our argument that there exists a *group law* that we will loosely call "addition" for rational points on elliptic curves, such that

1. Any two pairs of rational points can be "added" together to make another rational point on the curve

2. The point at infinity, $\mathcal{O}$ is always “on” the elliptic curve.
3. For any point $P \in E$ on the curve, there exists a unique $Q \in E$ such that $P + Q = \mathcal{O}$.
(Essentially an inverse)
4. This “addition” is an associative product.

Now, to give a mental model for what this should look like, consider the following. *Draw out the example below.*

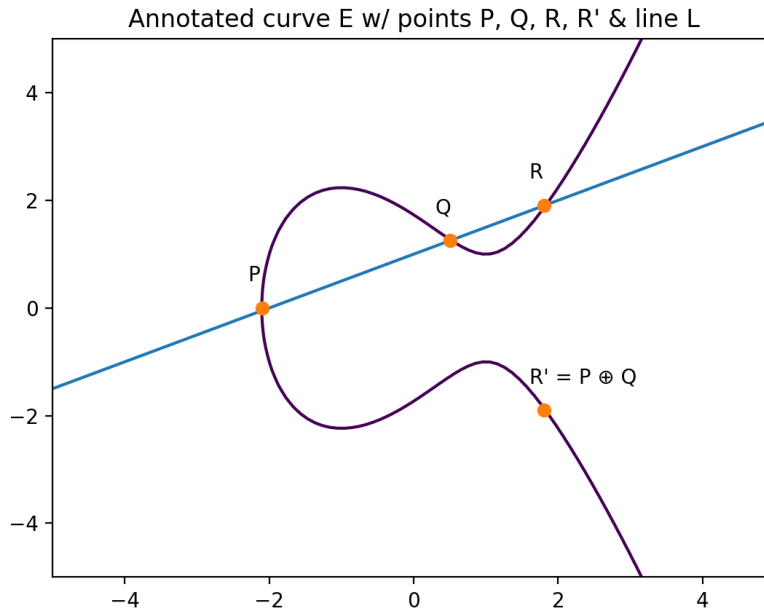


Figure 5: Group Law on Elliptic Curves

Group Law Formulas on E: $y^2 = x^3 + ax + b$

Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $P + Q = (x_3, y_3)$.

Case 1: $P \neq Q$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$x_3 = m^2 - x_1 - x_2$$

$$y_3 = m(x_1 - x_3) - y_1$$

$$P + Q = (x_3, y_3)$$

Case 2: $P = Q$ (Doubling)

$$m = \frac{3x_1^2 + a}{2y_1}$$

$$x_3 = m^2 - 2x_1$$

$$y_3 = m(x_1 - x_3) - y_1$$

$$2P = (x_3, y_3)$$

Key exposition: This is in fact the particular case that we are interested in for our correspondence between the incidence correspondence E , and our elliptic curve E . One can imagine that our Poncelet porism is the case where we pick a tangent line identified by the point P , and repeatedly add P to itself, resulting in $2P$, $3P$, etc... on E .

On an elliptic curve E , if there exists some integer $q \in \mathbb{Z}$ such that $qP = \mathcal{O}$, we say that P is a *torsion* point of E . Moreover, we say that

$$\text{Torsion}(E) = \{P \in E \mid qP = \mathcal{O} \text{ for some } q \in \mathbb{Z}\}.$$

Ex. (3-torsion)

$$y^2 = x^3 - 1$$

has $\text{Torsion}(E) = \{(0, i)\}$, since

$$\begin{aligned} 2(0, i)_m &= \frac{3 * 0^2 + 0}{2i} = 0 \\ 2(0, i)_x &= 0^2 - 2 * 0 = 0 \\ 2(0, i)_y &= 0(0 - 0) - i = -i \end{aligned}$$

has $2(0, i) = (0, -i) \in E$, and

$$((0, i) + (0, -i))_m = \infty$$

has $3(0, i) = \mathcal{O}$. So $y^2 = x^3 - 1$ has torsion of order 3.

Since we have already established our incidence correspondence E on C_1 and C_2^* , and we have taken (at face value from the Abel-Jacobi map) that E is a genus one Riemann surface, isomorphic to an elliptic curve $J(E)$, the solution to Poncelet's porism is now based on the following.

1. If $\text{Torsion}(J(E)) = \emptyset$, then there are no polygons in-and-circumscribed on C_1 and C_2 .
2. If $\text{Torsion}(J(E)) \neq \emptyset$, then there are infinitely many polygons in-and-circumscribed on C_1 and C_2 .

Both of these statements are independent of the choice of starting point, P_0 . Thus, Poncelet's Porism is (in an incredibly loose sense) proved.

□